

Michael Rizzo-Lopez

(previously Rizzo Smith)

Curriculum Vitae

Nashville, TN — michael.smith.4@vanderbilt.edu — mjosmith.github.io

EDUCATION

Ph.D Astrophysics

Vanderbilt University

Department of Physics and Astronomy

Advisor: Kelly Holley-Bockelmann

Aug. 2023 - Exp. 2029

Nashville, TN

Bachelor of Science, Physics and Astronomy & Planetary Science

Stony Brook University

Department of Physics and Astronomy

Magna Cum Laude with Honors in Physics

Aug. 2017 - May 2021

Stony Brook, NY

AWARDS AND FELLOWSHIPS

NSF Graduate Research Fellow (\$150,000)

Jun. 2025

Vanderbilt Graduate School Travel Grant

July 2024

McMinn Research Award

May 2024

Vanderbilt Robert T. Lagemann Award

Apr. 2024

Departmental Honors, Dept. of Physics

May 2021

Stony Brook Campus Scholarship (\$26,000)

Aug. 2017 - May 2021

PROFESSIONAL WORK AND AFFILIATIONS

NASA XRP 2026 Review Panel, Executive Secretary

Mar. 2026

LISA Consortium, Core Member

May 2025 - Present

Vanderbilt's Graduate Student Astrophysics Journal Research Club,
Co-Coordinator

Jan. 2025 - Present

Rebuilt existing club structure to emphasize practical presentation skills and implemented
a formal faculty feedback structure.

LISA Disc-IMRI Project Astrophysics Working Group, Code Captain

Dec. 2023 - Dec. 2025

Establishing Multimessenger Astronomy Inclusive Training (EMIT),

Sep. 2023 - Present

Graduate Student Scholar

ASAS-SN Project, Data Analyst

Jun. 2021 - July 2023

Research Assistant for the All Sky Automated Survey for Supernovae
at The Ohio State University.

OUTREACH

Astronomy on Tap Nashville, Host

2025 - Present

10-Years of Gravitational Waves, Vanderbilt Community Celebration, Poster

Sep. 2025

"It's OK to Be Gassy: How Cosmic Gas Guides Black Hole Collisions"

NASA Advocacy Day, Vanderbilt Federal Relations

July 2025

Met with congressional staff in Washington D.C. to emphasize funding public science and NASA.

NSF GRFP: Navigating the Application Panel, Panelist

Oct. 2025

Panel to prospective applicants.

Vanderbilt Society of Physics Students, Invited Panelist

Oct. 2025

Panel on graduate school application process.

EMIT Outreach, Invited Speaker

Jan. 2025

Career talk on computational astrophysics and higher ed.

Vanderbilt Society of Physics Students, Invited Panelist

Oct. 2024

Panel on graduate school application process.

Valor Collegiate Academy, Invited Speaker

Mar. 2024

Career talk to first-year class.

TEACHING

Graduate Teaching Assistant

Fall 2023 - Spring 2025

Course Title: ASTR 1010L: Intro to Astronomy Lab

Number of Students: ~21 per term

Private Tutor

Feb. 2020 - Oct. 2020

Intro Physics for Medical Professionals

Undergraduate Teaching Assistant

Spring 2020

Course Title: AST 203: Astronomy

PRESENTATIONS

Contributed Talk, Vanderbilt's Astrophysics Journal Research Club

Nov. 2025

Presenting: "*Relativistic Binary Precession: Impact on Eccentric Massive Binary Black Hole Accretion and Hydrodynamics*"

Contributed Talk, Vanderbilt's Astrophysics Journal Research Club

Jan. 2025

Presenting: "*The dynamics and electromagnetic signatures of accretion in unequal mass binary black hole inspirals*"

Contributed Talk, The 15th International LISA Symposium

July 2024

Presenting on behalf of the Disc-IMRI Code Comparison LISA Astrophysics Working Group

PUBLICATIONS

Reviewed Articles

(1 first-author, 5 co-author, 141 citations)

1. Andrea Derdzinski, Alexander J. Dittmann, Alessia Franchini, Alessandro Lupi, Noé Brucy, Pedro R. Capelo, Frédéric S. Masset, Raphaël Mignon-Risse, **Michael Rizzo Smith**, Edwin Santiago-Leandro, Martina Toscani, David A. Velasco-Romero, Robert Wissing, Mudit Garg, Lucio Mayer, Roberto Serafinelli, Lazaros Souvatzis, Daniel J. D'Orazio, and Jonathan Menu. "The LISA Astrophysics "Disc-IMRI" Code Comparison Project: Intermediate-Mass-Ratio Binaries in AGN-Like Discs". In: *arXiv e-prints*, arXiv:2512.10893 (Dec. 2025), arXiv:2512.10893. arXiv: 2512.10893 [astro-ph.HE]
2. Kelly Holley-Bockelmann, Fazeel Khan, Isaiah Williams, Jaelyn Roth, **Michael Rizzo Smith**, Kaitlin Porter, Jillian Bellovary, Andrea Derdzinski, and Andrea Macciò. "Handy Relation Between Binary Black Hole Merger Times and Host Galaxy Properties". In: *arXiv e-prints*, arXiv:2508.14253 (Aug. 2025), arXiv:2508.14253. arXiv: 2508.14253 [astro-ph.GA]
3. J. M. M. Neustadt, C. S. Kochanek, and **Rizzo Smith, M.** "Constraints on pre-SN outbursts from the progenitor of SN 2023ixf using the large binocular telescope". In: *MNRAS* 527.3 (Jan. 2024), pp. 5366–5373. arXiv: 2306.06162 [astro-ph.HE]
4. Matthew Kenworthy, [20 authors], and **Rizzo Smith, Michael**. "A planetary collision afterglow and transit of the resultant debris cloud". In: *Nature* 622.7982 (Oct. 2023), pp. 251–254. arXiv: 2310.08360 [astro-ph.EP]
5. **Rizzo Smith, M.**, C. S. Kochanek, and J. M. M. Neustadt. "The late time optical evolution of twelve core-collapse supernovae: detection of normal stellar winds". In: *MNRAS* 523.1 (July 2023), pp. 1474–1495. arXiv: 2212.09763 [astro-ph.HE]
6. A. Kawash, [11 authors], **Rizzo Smith, M.**, T. W. -S. Holoién, J. L. Prieto, and T. A. Thompson. "The Galactic Nova Rate: Estimates from the ASAS-SN and Gaia Surveys". In: *The Astrophysical Journal* 937.2, 64 (Oct. 2022), p. 64. arXiv: 2206.14132

Non-Reviewed Articles

(6 first-author, 13 citations)

1. M. Rizzo Smith and ASASSN-Team. "An Update on ASASSN-21qj: A Rapidly Fading, Sun-Like Star; Back With a Vengeance". In: *The Astronomer's Telegram* 15531 (July 2022), p. 1
2. M. Rizzo Smith and ASASSN-Team. "ASASSN-22el: A Deep Eclipse Event". In: *The Astronomer's Telegram* 15308 (Apr. 2022), p. 1
3. M. Rizzo Smith and ASASSN-Team. "ASASSN-21sa: A Deep Dimming Event". In: *The Astronomer's Telegram* 14937 (Sept. 2021), p. 1
4. M. Rizzo Smith and ASASSN-Team. "ASASSN-21qj: A Rapidly Fading, Sun-Like Star". In: *The Astronomer's Telegram* 14879 (Aug. 2021), p. 1
5. M. Rizzo Smith and ASASSN-Team. "ASASSN-21nn: An Unusual Dimming Event in a Red Giant Star". In: *The Astronomer's Telegram* 14803 (July 2021), p. 1
6. M. Rizzo Smith and ASASSN-Team. "ASASSN-21ml: Discovery of an Extreme ($\Delta g > 10$ mag) L-Dwarf Flare". In: *The Astronomer's Telegram* 14778 (July 2021), p. 1